

Patrícia S. Silva

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Patrícia is an environmental engineer with a PhD in Geophysics who specializes in fire–climate dynamics in tropical ecosystems. Her research examines how fire interacts with climate, vegetation, and human activities, with a particular focus on how climate change is reshaping present and future fire regimes. Her current research focuses on the climatic conditions that give rise to extreme, catastrophic fires and mapping ecosystem vulnerability. She integrates remote sensing, observational datasets, and climate models to understand recent changes in fire regimes and to project how they may evolve under future climate scenarios. She also contributes to work on extreme climate events (heatwaves and drought), climate modelling, and integrated fire management across South America.

EDUCATION

PhD in Geophysical Sciences and Geoinformation , <i>University of Lisbon, Portugal</i> Postgraduate Program in Meteorology , <i>Federal University of Rio de Janeiro, Brazil</i> A jointly supervised PhD program (via an international <i>cotutelle</i> agreement). Thesis entitled “Climate and vegetation dynamics and its impact on present and future fire regimes in Brazil”.	2019 – 2024
MSc in Energy and Environmental Engineering , <i>University of Lisbon, Portugal</i> Integrated Master’s Program at the Faculty of Sciences, including an ERASMUS Student Mobility for Studies Program in 2016 at the University of Groningen, The Netherlands.	2012 – 2018

PROFESSIONAL EXPERIENCE

Postdoctoral Associate , <i>Yale School of the Environment, Yale University, United States</i> Research topic: find climatic thresholds for wildfires in the tropical ecosystems of South America and model extreme fire occurrence.	2024 – present
Consultant , <i>ConservationXLabs, United States</i> Role: Data analysis, scientific and strategic advisory in the scope of an international collaborative project on fire in South American ecosystems.	Jan–Apr 2026
Consultant , <i>Wetlands International, Brazil</i> Role: Data analysis and scientific advisory in the scope of a project aimed at reducing carbon emissions in a natural heritage reserve in the Pantanal wetlands.	Aug 2021 – Jun 2022
Research Fellow , <i>Instituto Dom Luiz, University of Lisbon, Portugal</i> Research topic: understand the climatic and anthropogenic drivers of fire occurrence in the tropical savannas of Brazil (Cerrado).	2016 – 2019
Research Fellow , <i>Centre for Isotope Research, University of Groningen, The Netherlands</i> Research topic: quantifying and modelling methane emissions from the Groningen Gas Field using atmospheric observations.	2017
Scientific Animator , <i>Science4you, Portugal</i>	2014 – 2016

AWARDS, GRANTS AND FELLOWSHIPS

1st Place Poster Award at the Yale Climate Day & YCNCC Spring 2025 Symposium , <i>Yale Center for Natural Carbon Capture (YCNCC) and Yale Institute for Biospheric Studies (YIBS)</i>	2025
Best Doctoral Thesis in Meteorology , <i>Portuguese Association of Meteorology and Geophysics (APMG)</i>	2025
Outstanding Doctoral Thesis 2024 in Earth Sciences , <i>FCiências.ID (Faculty of Sciences of the University of Lisbon)</i>	2025
Early Career Scientist's Travel Support , <i>European Geosciences Union (EGU)</i>	2022
Training School Financial Support , <i>COST (European Cooperation in Science and Technology) Action DAMOCLES</i>	2022
Third best paper in the Student Paper Competition , <i>Latin American GRSS & ISPRS Remote Sensing Conference</i>	2020
PhD Fellowship , <i>Portuguese Foundation for Science and Technology (FCT)</i>	2019
Travel Grant , <i>American Geophysical Union (AGU) Natural Hazards Focus Group</i>	2016
One of the best works presented at VISIC Simpósio Internacional de Climatologia , <i>Brazilian Society of Meteorology (SBMET)</i>	2015

TEACHING AND MENTORING

Teaching Assistant:

Remote Sensing, <i>Faculty of Sciences, University of Lisbon</i>	2019
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Students Mentored:

Julianne Soares (Undergraduate), <i>Federal University of Rio de Janeiro</i>	2022 – 2025
Maria Mogrovejo Muller (Undergraduate), <i>Federal University of Rio de Janeiro</i>	2022 – 2023

Certifications:

- Environmental Leadership and Mentoring, Yale University.
- Certificate of Pedagogical Competencies (Government of Portugal; Certificate n.º F648014/2016).

Guest Lectures: Managing Ecosystems for Climate Change Solutions, *Yale University* (2025); Fire in the Earth System, *Federal University of Rio de Janeiro* (2025); Statistics, *Federal University of Rio de Janeiro* (2024 and 2025); Climatology, *Federal University of Rio de Janeiro* (2021).

Invited Talks and Roundtables: Wildfire in Savanna Ecosystems II — Case Studies and Considerations, *Global Wildfire Collective* (2026); NYC Climate Week Technical Roundtable, *Global Wildfire Collective* (2026); Firethon, *Amazon Cooperation Treaty Organization* (2026); Climate Detectives, *European Space Education Resource Office* (2025); Course on Satellite-Based Fire Monitoring in Pantanal, *Federal University of Rio de Janeiro* (2022); Training Course on Satellite-Based Fire Monitoring in Cerrado, *Federal University of Rio de Janeiro* (2021).

Course development and design: Compreendendo o fogo no Pantanal por meio do monitoramento por satélite (EN: Understanding fire in the Pantanal through remote sensing), *Wetlands International* (2025).

H-index: 10 (Google Scholar); **Total Citations:** 873 (Google Scholar). **Published papers:** 14 (7 as first-author).

Submitted or under review:

- Dutra, A.K.N., Carvalho, L.A., Silva, P.S., Belém, L.B., Rodrigues, A., Schmidt, I.B., Roque, F., Veiga, R., Getirana, A., Libonati, R. **Fire as an overlooked threat to water resources in the Global South.**
- Burton, C., Di Giuseppe, F., Jones, M.W., Kelley, D.I., Barbosa, M.L., McNorton, J., Potts, R., Segura-Garcia, C., Abatzoglou, J.T., Alin Rosu, I. Andela, N., ... , Silva, P.S., ..., Turco, M., van der Werf, G.R., van Wees, D., Veraverbeke, S., Wessel, J.B., and Zheng, B. 2026. **State of Wildfires 2025–2026.** Earth System Science Data Discussions, 2026, pp.1-111. <https://doi.org/10.5194/essd-2026-587>
- Silva, P.S., Macedo, M., Morton, D., Alencar, A., Randerson, J.T., Starinchak, B., Brando, P.M. 2026. **Climatic vulnerability of the Amazon to large fires.**

Published:

- Barlow, J., Carvalho, N.S., Nunes, C.A., Aguiar, A.P.D., Alencar, A., Anderson, L.O., Aragão, L.E., Baccaro, F., Barrett, M., Berenguer, E., Bodolai, K., Brando, P.M., ..., Silva, P.S., Silveira, J.M., Silvério, D.V., Sitch, S., Tavares, P.A., Telhado, C., Vieira, I.C.G. & Wiederhecker, H.C. 2026. **Rapid increase of climate extremes reveals new areas of concern in Amazonia.** Communications Earth & Environment, 7(1), p.746. <https://doi.org/10.1038/s43247-026-03975-1>
- Borges, S.L., dos Santos, A.C., Silva, P.S. and Schmidt, I.B., 2026. **Fire as a productive and preventive technology: traditional knowledge and risk governance in Kalunga agropastoral mosaics.** Fire Ecology, 22(1), p.95. <https://doi.org/10.1186/s42408-026-00522-3>
- Belém, L.B., Albuquerque, R., Peres, L.F., Santos, F.L., Rodrigues, J.A., Menezes, L.S., Silva, P.S., Trigo, R.M., DaCamara, C.C., Russo, A. and Geirinhas, J., 2026. Dry–Hot Compound Events Driving the 2024 Pantanal Wildfires. **International Journal of Climatology**, p.e70312. <https://doi.org/10.1002/joc.70312>
- Silva, P.S., Libonati, R., Gonçalves, L.G. and DaCamara, C.C., 2025. The climatic patterns that control regional fire activity in the Brazilian savanna. **Agricultural and Forest Meteorology**, 374, p.110792. <https://doi.org/10.1016/j.agrformet.2025.110792>
- Kelley, D.I., Burton, C., Di Giuseppe, F., Jones, M.W., Barbosa, M.L., Brambleby, E., McNorton, J.R., Liu, Z., Bradley, A.S., Blackford, K., Burke, E., ..., Silva, P.S., ... Kolden, C.A., 2025. State of Wildfires 2024–2025. **Earth System Science Data**, 17(10), pp.5377-5488. <https://doi.org/10.5194/essd-17-5377-2025>
- Anand, M., Hamed, R., Linscheid, N., Silva, P.S., Andre, J., Zscheischler, J., Garry., F.K., Bastos, A., 2024. Winter climate preconditioning of summer vegetation extremes in the Northern Hemisphere. **Environmental Research Letters**, 19(094045). <https://doi.org/10.1088/1748-9326/ad627d>
- Silva, P.S., Rodrigues, J.A., Nogueira, J., Moura, L.C., Enout, A., Cuiabália, C., DaCamara, C.C., Pereira, A.A., Libonati, R., 2024. Joining forces to fight wildfires: Science and management in a protected area of Pantanal, Brazil. **Environmental Science & Policy**, 159(103818). <https://doi.org/10.1016/j.envsci.2024.103818>
- Silva, P.S., Geirinhas, J.L., Lapere, R., Laura, W., Cassain, D., Alegría, A. and Campbell, J., 2022. Heatwaves and fire in Pantanal: Historical and future perspectives from CORDEX-CORE. **Journal of**

Environmental Management, 323, p.116193.

<https://doi.org/10.1016/j.jenvman.2022.116193>

- Libonati, R., Geirinhas, J.L., Silva, P.S., Monteiro dos Santos, D., Rodrigues, J.A., Russo, A., Peres, L.F., Narcizo, L., Gomes, M.E., Rodrigues, A.P. and DaCamara, C.C., 2022. Drought–heatwave nexus in Brazil and related impacts on health and fires: A comprehensive review. **Annals of the New York Academy of Sciences**, 1517(1), p.44-62.
<https://doi.org/10.1111/nyas.14887>
- Libonati, R., Geirinhas, J.L., Silva, P.S., Russo, A., Rodrigues, J.A., Belém, L.B., Nogueira, J., Roque, F.O., DaCamara, C.C., Nunes, A.M. and Marengo, J.A., 2022. Assessing the role of compound drought and heatwave events on unprecedented 2020 wildfires in the Pantanal. **Environmental Research Letters**, 17(1), p.015005.
<https://doi.org/10.1088/1748-9326/ac462e>
- Silva, P.S., Nogueira, J., Rodrigues, J.A., Santos, F.L., Pereira, J.M., DaCamara, C.C., Daldegan, G.A., Pereira, A.A., Peres, L.F., Schmidt, I.B. and Libonati, R., 2021. Putting fire on the map of Brazilian savanna ecoregions. **Journal of Environmental Management**, 296, p.113098.
<https://doi.org/10.1016/j.jenvman.2021.113098>
- Silva, P.S., Rodrigues, J.A., Santos, F.L.M., Pereira, A.A., Nogueira, J., DaCamara, C.C. and Libonati, R., 2020, March. Drivers of burned area patterns in Cerrado: the case of MATOPIBA region. In: 2020 IEEE Latin American GRSS & ISPRS Remote Sensing Conference (LAGIRS) (pp. 542-547). **IEEE**.
<https://doi.org/10.1109/LAGIRS48042.2020.9165665>
- Silva, P.S., Bastos, A., Libonati, R., Rodrigues, J.A. and DaCamara, C.C., 2019. Impacts of the 1.5°C global warming target on future burned area in the Brazilian Cerrado. **Forest Ecology and Management**, 446, pp.193-203. <https://doi.org/10.1016/j.foreco.2019.05.047>
- Silva, P., Bastos, A., DaCamara, C.C. and Libonati, R., 2016. Future projections of fire occurrence in Brazil using EC-Earth climate model. **Revista Brasileira de Meteorologia**, 31, pp.288-297.
<https://dx.doi.org/10.1590/0102-778631320150142>

BOOK CHAPTERS

- Silva, P.S., Libonati, R., Marengo, J.A., Costa, M.C., Alves, L.M., Schmidt, I.B. (2025). Fire in the Anthropocene. In: Fidelis, A., Pivello, V.R. (eds) Fire in the South American Ecosystems. Ecological Studies, vol 250. **Springer**, Cham. https://doi.org/10.1007/978-3-031-89372-8_13
- Silva, P.S., Libonati, R., Schmidt, I.B., Nogueira, J., DaCamara, C.C., 2024. Climate Change and Fire: The Case of Cerrado, the Brazilian Savanna. In: Mishra, M., de Lucena, A.J., Maharaj, B. (eds) Climate Change and Regional Socio-Economic Systems in the Global South. **Springer**, Singapore.
https://doi.org/10.1007/978-981-97-3870-0_6

REPORTS AND EDUCATIONAL MATERIALS

- Damasceno-Junior, G. A., Guerra, A., de Matos Martins Pereira, A., Berlinck, C. N., de Oliveira Roque, F., Ebert, A., Rocha, A. R., Nunes, A. V., Pott, A., Oliveira, B., da Cunha, C. N., Ribeiro, D. B., Bolzan, F. P., Bao, F., Fernandes, G. W., Anderson, L. O., Oliveira, M., Silva, P., Libonati, R., da Silva, R. H., and Santos,

S. (2024). Manejo Integrado do Fogo no Pantanal - Um roteiro para o fogo bom.

<http://dx.doi.org/10.13140/RG.2.2.34323.25121>

- Araújo, Y., de Oliveira Pismel, G., dos Reis, J. B. C., and Anderson, L. O. (2024). É Fogo! Pantanal: guia de atividades. Ed. dos Autores, São José dos Campos, SP. <https://efogo.com.br>
- United Nations Environment Programme, 2022. Spreading like Wildfire – The Rising Threat of Extraordinary Landscape Fires. A **UNEP Rapid Response Assessment**. Nairobi. <https://www.unep.org/resources/report/spreading-wildfire-rising-threat-extraordinary-landscape-fires>

CONFERENCES

Total (including co-authored): 15 oral presentations, 13 poster presentations and 1 multimedia format (vPico).

As presenting author: 8 oral presentations and 6 poster presentations.

Selected conference participations:

- [Silva, P.S.](#), Brando, P.M. **Climatic thresholds for Amazon wildfires: historical and future perspectives**. 2026. Coupled Model Intercomparison Project (CMIP) Community Workshop 2026, 9-13 March, Kyoto, Japan. (poster)
- [Silva, P.S.](#), Brando, P.M. Under fire: the climatic drivers of extreme fires in the Amazon. 2025. **11th International Fire Ecology and Management Congress**, 2–6 December, New Orleans, United States of America. (oral)
- [Silva, P.S.](#), Libonati, R., DaCamara, C.C., Brando, P.M. 2025. The burning case of the Brazilian savannas: Unveiling regional fire patterns and drivers. **61st Annual Meeting of the Association for Tropical Biology and Conservation**, 29 June–4 July, Oaxaca, Mexico. (oral)
- [Silva, P.S.](#), Geirinhas, J.L., Rodrigues, J.A., DaCamara, C.C., Libonati, R. 2023. Compound drought-heatwave conditions: a new threat for the fire-prone Cerrado? **8th International Wildland Fire Conference (WILDFIRE 2023)**, 16–19 May, Porto, Portugal. (poster)
- [Silva P.](#), Bastos A., DaCamara C., Libonati R. 2019. The 1.5°C global warming target and its impact on future burned area in the Brazilian savanna. **EGU General Assembly 2019**, 7–12 April, Vienna, Austria. (oral)
- [Silva P.](#), Bastos A., DaCamara C., Libonati R. 2016. Trends in fire risk and burned area in Brazil in the 20th century. **AGU Fall Meeting 2016**, 12–16 December, San Francisco, California, USA. (poster)

TRAINING SCHOOLS AND WORKSHOPS

Coupled Model Intercomparison Project (CMIP) Community Workshop 2026 , <i>World Climate Research Programme (WCRP)</i> [Website]	2026
Forest resilience in climate risk assessments of tropical forests: current knowledge and future research , <i>Yale University</i>	2025
DAMOCLES Training School on Compound Events , <i>Eötvös Loránd University (ELTE)</i> [Website] Resulted in a scientific publication (Anand et al., 2024).	2022

Pyrolife Training Event - Understanding fire human dimensions and planning , <i>Fundación de Ecología del Fuego y Gestión de Incendios Pau Costa</i> [Website]	2021
CORDEX Central America and South America Online Paper-Writing Workshop on Regional Climate Modeling , <i>World Climate Research Programme (WCRP)</i> [Website] Resulted in a scientific publication (Silva et al., 2022).	2020
Training Workshop on Diagenic Models and ESMs of Intermediate Complexity , <i>Université Libre de Bruxelles</i> [Website]	2017
Training Workshop on Regional to Global Modelling , <i>Centre National de la Recherche Scientifique - Institut Pierre Simon Laplace (CNRS-IPSL)</i> [Website]	2017

RESEARCH PROJECTS

Improving Amazon fire risk forecasts to minimize impacts on tropical forest ecosystems and human health
Role: Postdoctoral Researcher; Funding: National Aeronautics and Space Administration (NASA)

Rede Biota Cerrado: Projecto Asociado Manejo Integrado do Fogo (MIF) [[Website](#)]

Role: Participant Researcher; Funding: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq)

OVERHEAT.SA - COLlaboratiVE Research on Compound Drought and HEATWave events in South America

Role: Participant Researcher; Funding: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) [443285/2023-3]

BrFLAS - Brazilian Fire-Land-Atmosphere System [[Website](#)]

Role: Graduate Research Fellow; Funding: Fundação de Amparo à Pesquisa do Estado de São Paulo (FAPESP) [FAPESP/1389/2014]

ANDURÁ - Uso de sensoriamento remoto para avaliação da qualidade de programas de fogo prescrito

Role: Participant Researcher; Funding: Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq) [441971/2018-0]

SCIENCE COMMUNICATION, OUTREACH AND PEER-REVIEW

Outreach activities:

Climate Detectives - European Space Agency (ESA) [Website]	2024 – 2025
PubhD Lisbon [Website]	2023
Cartas com Ciência (EN: Letters with Science) [Website]	2021 – 2023

Scientific networks and initiatives:

Science Team - The State of Wildfires Project [Website]	2025 – present
European Network on Extreme fire behaviOr (NERO) COST Action CA22164 [Website]	2025 – present

Peer-review:

Review Editor in Frontiers in Forests and Global Change - Fire and Forests	2022 – present
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Invited Peer Referee: Global Change Biology, Frontiers in Forests and Global Change, Anuário do Instituto de Geociências, Fire Ecology, Natural Hazards and Earth System Sciences, Biological Conservation, Resources, Environment and Sustainability.

HIGHLIGHTED SCIENTIFIC SKILLS

Technical skills:

- Programming: Python, MATLAB, R, C++, Bash, Climate Data Operators (CDO), netCDF Operators (NCO)
- Development: HTML/CSS
- Software: ArcGIS/QGIS, LaTeX, AutoCAD

Certifications:

- S130/S190/L180 Basic Wildland Firefighter Training, National Wildfire Coordinating Group (NWCG).

Languages: fluent in Portuguese and English (Cambridge C1 Advanced: Grade A); elementary proficiency in Spanish, French and Chinese (HSK1).